2. Requirement Specification and Analysis

2.1 Epics

E1: User Authentication and Profile Management

Description: As a farmer or administrator, I want to securely log in, register, and manage my profile so that my data and farm configurations remain private and secure.

E2: Real-Time Sensor Data Monitoring

Description: As a farmer, I want to monitor real-time environmental data (soil moisture, temperature, humidity) from my fields so that I can make informed decisions about crop care.

E3: Intelligent Irrigation Control

Description: As a farmer, I want to control irrigation systems manually or set automatic triggers based on sensor thresholds so that water usage is optimized and crops receive adequate hydration.

E4: AI-Driven Crop Recommendations

Description: As a farmer, I want to receive AI-generated crop recommendations based on soil and weather analysis so that I can maximize yield and select the most suitable crops for the season.

E5: Alerting and Notification System

Description: As a farmer, I want to receive immediate alerts about critical conditions (e.g., low soil moisture, sensor failure) so that I can take timely corrective actions to prevent crop damage.

E6: Historical Data Analytics and Visualization

Description: As a farmer, I want to view historical trends and graphical visualizations of farm data so that I can analyze long-term patterns and improve farming strategies.

2.2 User Stories

Epic E1: User Authentication and Profile Management

E1-US1: User Registration

Description: As a new user, I want to create an account using my email and phone number so that I can access the system features.

Acceptance Criteria:

Given the user is on the registration screen,

When they enter valid name, email, phone, and password,

Then the system should create a new account and redirect to the login page.

E1-US2: Secure Login

Description: As a registered user, I want to log in using my credentials so that I can access my personal dashboard.

Acceptance Criteria:

Given the user has a valid account,

When they enter the correct email and password,

Then the system should authenticate the user and grant access to the dashboard.

E1-US3: Profile Management

Description: As a user, I want to update my personal information and change my password so that my account details remain current and secure.

Acceptance Criteria:

Given the user is logged in,

When they navigate to the profile settings and update their phone number,

Then the system should save the changes and display a success message.

Epic E2: Real-Time Sensor Data Monitoring

E2-US1: View Field Overview

Description: As a farmer, I want to see a list of all my fields with their current status so that I can quickly assess the overall farm health.

Acceptance Criteria:

Given the user is on the dashboard,

When the page loads,

Then the system should display all registered fields with summary indicators (e.g., "Healthy", "Needs Water").

E2-US2: Detailed Sensor Readings

Description: As a farmer, I want to view specific readings from individual sensors (e.g., Sensor ID 14) so that I can pinpoint issues in specific areas.

Acceptance Criteria:

Given the user selects a specific field,

When they tap on a sensor node,

Then the system should show the latest values for soil moisture, temperature, and humidity.

E2-US3: Sensor Status Monitoring

Description: As a farmer, I want to know if a sensor is offline or has low battery so that I can perform maintenance.

Acceptance Criteria:

Given a sensor has stopped sending data,

When the user views the sensor list,

Then the system should display an "Offline" status indicator next to that sensor.

Epic E3: Intelligent Irrigation Control

E3-US1: Manual Irrigation Toggle

Description: As a farmer, I want to remotely turn the water pump on or off so that I can irrigate my field immediately when needed.

Acceptance Criteria:

Given the irrigation system is connected,

When the user presses the "Start Irrigation" button,

Then the system should send a command to the actuator and update the status to "Irrigating".

E3-US2: Threshold-Based Automation

Description: As a farmer, I want to set a soil moisture threshold (e.g., 30%) so that the system automatically irrigates when the soil gets too dry.

Acceptance Criteria:

Given the automation mode is enabled,

When the soil moisture drops below the defined threshold,

Then the system should automatically trigger the irrigation pump without user intervention.

E3-US3: Irrigation Logging

Description: As a farmer, I want to see a history of when irrigation occurred so that I can track water usage.

Acceptance Criteria:

Given an irrigation cycle has completed,

When the user views the irrigation logs,

Then the system should list the start time, end time, and total duration of the event.

Epic E4: AI-Driven Crop Recommendations

E4-US1: Request Crop Recommendation

Description: As a farmer, I want to request a crop recommendation based on my field's current soil data so that I can plant the most viable crop.

Acceptance Criteria:

Given the user is on the recommendation screen,

When they click "Analyze Field",

Then the system should process the soil parameters and display a recommended crop (e.g., "Wheat").

E4-US2: View Recommendation Confidence

Description: As a farmer, I want to see the confidence score of the AI prediction so that I can trust the recommendation.

Acceptance Criteria:

Given a recommendation is generated,

When the result is displayed,

Then the system should show a percentage score (e.g., "85% Confidence").

E4-US3: Accept Recommendation

Description: As a farmer, I want to mark a recommendation as "Accepted" so that the system tracks what I decided to plant.

Acceptance Criteria:

Given a recommendation is displayed,

When the user clicks "Accept",

Then the field's current crop status should be updated to match the recommendation.

Epic E5: Alerting and Notification System

E5-US1: Critical Threshold Alerts

Description: As a farmer, I want to receive a notification when soil moisture is critically low so that crops do not die.

Acceptance Criteria:

Given the soil moisture is below the critical limit,

When the sensor transmits the data,

Then the system should generate a "Critical Alert" and notify the user.

E5-US2: View Unread Alerts

Description: As a farmer, I want to see a badge count of unread alerts so that I know if I missed important information.

Acceptance Criteria:

Given there are new alerts,

When the user opens the app,

Then the notification icon should show a badge with the number of unread items.

E5-US3: Resolve Alerts

Description: As a farmer, I want to mark an alert as resolved so that I can clear my notification feed.

Acceptance Criteria:

Given an active alert exists,

When the user clicks "Mark as Resolved",

Then the alert should be moved to the history archive and the status updated.

Epic E6: Historical Data Analytics and Visualization

E6-US1: View Moisture Trends

Description: As a farmer, I want to see a line graph of soil moisture over the last week so that I can understand drying patterns.

Acceptance Criteria:

Given historical data exists,

When the user selects the "Weekly View" on the chart,

Then the system should render a line graph showing moisture levels over the past 7 days.

E6-US2: Export Data

Description: As a farmer, I want to export my farm data so that I can keep offline records.

Acceptance Criteria:

Given the user is on the analytics screen,

When they click "Export",

Then the system should generate a downloadable report (e.g., CSV or PDF).

E6-US3: Compare Fields

Description: As a farmer, I want to compare the water usage of two different fields so that I can identify inefficiencies.

Acceptance Criteria:

Given the user selects two fields,

When they choose "Compare",

Then the system should display side-by-side statistics for both fields.

2.3 Test Cases

E1: User Authentication

2.3.1. Test Case 1 This test case verifies the user registration process. It checks if the system correctly accepts valid user details (Name, Email, Phone, Password) and creates a new user record in the database, followed by a redirection to the login page.

Table 2-1 Verify User Registration with Valid Data

Test ID: TC-E1-US1-01 **US ID:** E1-US1

Test Case Description: Ensure a new user can register successfully with correct details.

Inputs: Name="Ali Khan", Email="ali@test.com", Phone="+923001234567", Password="Pass123"

Expected Result: Account created in `users` table; User redirected to Login screen.

2.3.2. Test Case 2 This test case validates the user login mechanism. It ensures that a registered user can access the dashboard by providing correct credentials, verifying that a JWT token is generated and the user is authenticated successfully.

Table 2-2 Verify Login with Valid Credentials

Test ID: TC-E1-US2-01 **US ID:** E1-US2

Test Case Description: Ensure a registered user can access the dashboard.

Inputs: Email="ali@test.com", Password="Pass123"

Expected Result: JWT token generated; User redirected to Dashboard; HTTP 200 OK.

E2: Real-Time Monitoring

2.3.3. Test Case 3 This test case ensures the accuracy of real-time sensor data display. It verifies that the latest readings for soil moisture, temperature, and humidity sent by the sensor are correctly reflected on the user interface.

Table 2-3 Verify Real-Time Data Display

Test ID: TC-E2-US2-01 **US ID:** E2-US2

Test Case Description: Check if the latest sensor reading is shown correctly on the UI.

Inputs: SensorID=14 sends {moisture: 45%, temp: 30°C}

Expected Result: UI displays "Moisture: 45%", "Temp: 30°C"; Data matches `sensor_readings` table.

E3: Irrigation Control

2.3.4. Test Case 4 This test case verifies the manual control of the irrigation system. It checks if the water pump can be successfully activated via the application's interface and if the status is updated in real-time.

Table 2-4 Verify Manual Pump Start

Test ID: TC-E3-US1-01	**US ID:** E3-US1
Test Case Description:	Ensure the pump turns on when the user clicks the start button.
Inputs:	Click "Start Irrigation" button for Field ID 6.
Expected Result:	API returns success; Pump status updates to "ON"; New entry in `irrigation_logs`.

2.3.5. Test Case 5 This test case validates the automatic irrigation logic. It simulates a low soil moisture condition to verify that the system automatically triggers the water pump and generates an appropriate alert without user intervention.

Table 2-5 Verify Automatic Trigger on Low Moisture

Test ID: TC-E3-US2-01	**US ID:** E3-US2
Test Case Description:	Ensure pump starts automatically when moisture is below threshold.
Inputs:	Set Threshold=30%; Simulate Sensor Reading=25%.
Expected Result:	System triggers irrigation command; Alert generated; Pump status "ON".

E4: AI Recommendations

2.3.6. Test Case 6 This test case tests the AI-driven crop recommendation feature. It verifies that the system can process field data (soil type, season) and return a valid crop suggestion along with a confidence score.

Table 2-6 Verify Crop Recommendation Generation

Test ID: TC-E4-US1-01	**US ID:** E4-US1
Test Case Description:	Ensure the AI model returns a valid crop suggestion.

Test ID: TC-E4-US1-01	**US ID:** E4-US1
Inputs:	Field ID=6 (Soil: Loamy, Season: Rabi)
Expected Result:	System returns "Wheat" with confidence score; Entry created in `crop_recommendations`.

E5: Alerts

2.3.7. Test Case 7 This test case verifies the critical alert system. It ensures that when a sensor reports a value below the critical threshold (e.g., extremely low soil moisture), the system generates a high-priority alert to notify the user.

Table 2-7 Verify Critical Alert Delivery

Test ID: TC-E5-US1-01	**US ID:** E5-US1
Test Case Description:	Ensure an alert is created when a sensor reports critical values.
Inputs:	Sensor reading: Moisture=10% (Critical limit < 15%)
Expected Result:	New record in `alerts` table; Notification badge increments on UI.

E6: Analytics

2.3.8. Test Case 8 This test case validates the data visualization capabilities. It checks if the system correctly renders historical soil moisture trends over the past week on a line graph, ensuring data points are accurate.

Table 2-8 Verify Weekly Trend Graph

Test ID: TC-E6-US1-01	**US ID:** E6-US1
Test Case Description:	Ensure the chart renders 7 data points for the last week.
Inputs:	Select "Last 7 Days" filter.
Expected Result:	Line chart displays with correct dates on X-axis and moisture values on Y-axis.

2.4 User Interface Design (Prototypes)

1. Login Screen

UI Layout: Centered card layout with logo at the top.

Fields and Components:

`Email Address` (Input field, email validation)

`Password` (Input field, obscured text)

`Login` (Primary Button)

`Register` (Text Link)

Validation Rules: Email must be valid format; Password cannot be empty.

Navigation Behavior: On success -> Redirect to Dashboard. On failure -> Show error toast.

2. Dashboard (Sensor Overview)

UI Layout: Grid layout of cards representing different fields.

Fields and Components:

`Field Name` (Header)

`Status Indicator` (Green/Red dot)

`Quick Stats` (Avg Moisture, Temp)

`Add Field` (Floating Action Button)

Validation Rules: N/A (Display only).

Navigation Behavior: Tap on a Field Card -> Navigate to Detailed Sensor Data Page.

3. Detailed Sensor Data Page

UI Layout: Tabbed view (Live Data, History, Settings).

Fields and Components:

`Gauge Chart` (Visualizing Soil Moisture %)

`Digital Readout` (Temperature, Humidity)

`Last Updated` (Timestamp)

`Sensor ID` (Label)

Validation Rules: Data refreshes every 30 seconds.

Navigation Behavior: Back button -> Dashboard.

4. AI Crop Recommendation Screen

UI Layout: Form input followed by result card.

Fields and Components:

`Soil Type` (Dropdown)

`Season` (Dropdown)

`Analyze` (Button)

`Recommendation Card` (Displays Crop Name, Confidence %, Yield Estimate)

`Accept Recommendation` (Button)

Validation Rules: All dropdowns must be selected before clicking Analyze.

Navigation Behavior: Accept -> Updates Field Details.

5. Irrigation Control Panel

UI Layout: Toggle switches and slider controls.

Fields and Components:

`Manual Control` (On/Off Switch)

`Auto-Irrigation` (Checkbox)

`Moisture Threshold` (Slider 0-100%)

`Status` (Text: "Pump Running" / "Idle")

Validation Rules: Cannot enable Manual Control if Auto-Irrigation is active (mutex).

Navigation Behavior: Toggle Switch -> Sends API request immediately.

6. Alerts & Notifications Screen

UI Layout: List view of notification cards.

Fields and Components:

`Alert Icon` (Warning triangle, Info circle)

`Title` (e.g., "Low Moisture Warning")

`Time` (Relative time, e.g., "2 mins ago")

`Mark as Read` (Swipe action or button)

Validation Rules: Unread items highlighted.

Navigation Behavior: Tap Alert -> Navigate to relevant Field/Sensor page.

7. Settings / Profile Page

UI Layout: List of settings categories.

Fields and Components:

`Profile Image` (Avatar)

`Name`, `Email` (Editable fields)

`Change Password` (Button)

`Notification Preferences` (Toggles)

`Logout` (Destructive Button)

Validation Rules: Password change requires old password verification.

Navigation Behavior: Logout -> Redirect to Login Screen.

2.5 Traceability Matrix

Epic	User Story	Test Case	UI Screen
E1 (Auth)	E1-US1 (Register)	TC-E1-US1-01	Login Screen (Register Tab)
E1 (Auth)	E1-US2 (Login)	TC-E1-US2-01	Login Screen
E1 (Auth)	E1-US3 (Profile)	N/A (Standard Update)	Settings / Profile Page
E2 (Monitoring)	E2-US1 (Field Overview)	N/A (Visual Check)	Dashboard
E2 (Monitoring)	E2-US2 (Sensor Detail)	TC-E2-US2-01	Detailed Sensor Data Page
E2 (Monitoring)	E2-US3 (Status)	N/A (Status Check)	Dashboard / Sensor Page
E3 (Irrigation)	E3-US1 (Manual)	TC-E3-US1-01	Irrigation Control Panel
E3 (Irrigation)	E3-US2 (Auto)	TC-E3-US2-01	Irrigation Control Panel

E3 (Irrigation)	E3-US3 (Logs)	N/A (Log Check)	Detailed Sensor Data Page (History)
E4 (AI)	E4-US1 (Request Rec)	TC-E4-US1-01	AI Crop Recommendation Screen
E4 (AI)	E4-US2 (Confidence)	N/A (Visual Check)	AI Crop Recommendation Screen
E4 (AI)	E4-US3 (Accept)	N/A (State Change)	AI Crop Recommendation Screen
E5 (Alerts)	E5-US1 (Critical)	TC-E5-US1-01	Alerts & Notifications Screen
E5 (Alerts)	E5-US2 (Unread)	N/A (Badge Check)	Dashboard (Icon)
E5 (Alerts)	E5-US3 (Resolve)	N/A (Action Check)	Alerts & Notifications Screen
E6 (Analytics)	E6-US1 (Trends)	TC-E6-US1-01	Detailed Sensor Data Page (History)
E6 (Analytics)	E6-US2 (Export)	N/A (Download)	Settings / Profile Page
E6 (Analytics)	E6-US3 (Compare)	N/A (Multi-view)	Dashboard